

Common Acids and Bases

Subject & topics: Chemistry | Physical | Acids & Bases **Stage & difficulty:** A Level P1

Test your knowledge of common acids and bases with the questions below.

Part A

Nitric acid

Enter the molecular formula for nitric acid (listing hydrogen in the formula first).

Part B

Sulfuric acid

Enter the molecular formula for sulfuric acid (listing hydrogen in the formula first).

Part C

Hydrochloric acid

Enter the molecular formula for hydrochloric acid (listing hydrogen in the formula first).

Part D

Carbonic acid

Enter the molecular formula for carbonic acid (listing hydrogen in the formula first).

Part E

Ammonia

Enter the molecular formula for ammonia.

Part F

Potassium hydroxide

Enter the molecular formula for potassium hydroxide.

Part G

Sodium hydroxide

Enter the molecular formula for sodium hydroxide.

Part H

Ethanoic acid

Ethanoic acid is an example of a carboxylic acid, containing the carboxylic acid group composed of four atoms. If not drawing out the organic structure, we still usually make the functional group it contains clear, by writing the formula as:

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Items:

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Created for isaacphysics.org by Andrea Chlebkova

Question deck:

STEM SMART Chemistry Week 21

Acid-Base Terminology

Subject & topics: Chemistry | Physical | Acids & Bases **Stage & difficulty:** A Level P1

The terminology surrounding acids and bases can be a bit confusing. Answer the questions below to test your understanding of this topic.

Part A

Fully dissociated

What do we call an acid or base that fully dissociates in aqueous solution?

Part B

High mol dm^{-3}

What do we call an acid or alkali solution that has a high value of mol dm^{-3} ?

Part C

Low mol dm^{-3}

What do we call an acid or alkali solution that has a low value of mol dm^{-3} ?

Part D

Acidic solutions

An acid with a very high K value is a acid, but it can still be if we create a solution of it with large amounts of water. At the same concentration, a acid will have a lower pH than a acid.

Items:

Part E

Two acidic protons

What do we call an acid such as H_2SO_4 that has two acidic protons it can lose in succession?

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Question deck:

STEM SMART Chemistry Week 21

Brønsted-Lowry & Lewis 4

Essential Pre-Uni Chemistry J1.4

Subject & topics: Chemistry | Physical | Acids & Bases

Stage & difficulty: A Level P1

Give the conjugate acid of NH_3 .

Question deck:

STEM SMART Chemistry Week 21

Brønsted-Lowry & Lewis 5

Essential Pre-Uni Chemistry J1.5

Subject & topics: Chemistry | Physical | Acids & Bases **Stage & difficulty:** A Level P1

Give the conjugate base of H_2SO_4 .

Question deck:

STEM SMART Chemistry Week 21

Brønsted-Lowry & Lewis 6

Essential Pre-Uni Chemistry J1.6

Subject & topics: Chemistry | Physical | Acids & Bases

Stage & difficulty: A Level P2

Give the conjugate base of NH_3 .

Question deck:

STEM SMART Chemistry Week 21

Brønsted-Lowry & Lewis 7

Essential Pre-Uni Chemistry J1.7

Subject & topics: Chemistry | Physical | Acids & Bases

Stage & difficulty: A Level P2

Give the conjugate base of $\text{H}_3\text{N}^+\text{CH}_2\text{COO}^-$.

Question deck:

STEM SMART Chemistry Week 21

Brønsted-Lowry & Lewis 8

Essential Pre-Uni Chemistry J1.8

Subject & topics: Chemistry | Physical | Acids & Bases

Stage & difficulty: A Level P2

Give the conjugate acid of $\text{H}_3\text{N}^+\text{CH}_2\text{COO}^-$.

Question deck:

STEM SMART Chemistry Week 21

Brønsted-Lowry & Lewis 9

Essential Pre-Uni Chemistry J1.9

Subject & topics: Chemistry | Physical | Acids & Bases **Stage & difficulty:** A Level P1

Give the conjugate acid of PO_4^{3-} .

Question deck:

STEM SMART Chemistry Week 21

Brønsted-Lowry & Lewis 10

Essential Pre-Uni Chemistry J1.10

Subject & topics: Chemistry | Physical | Acids & Bases

Stage & difficulty: A Level P1

Give the conjugate base of CH_3OH .

Question deck:

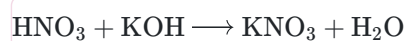
STEM SMART Chemistry Week 21

Acid-Base Reactions

Subject & topics: Chemistry | Physical | Acids & Bases **Stage & difficulty:** A Level P1

Part A

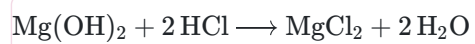
Reaction 1



- ☐ acid-base reaction
- ☐ other

Part B

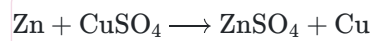
Reaction 2



- ☐ acid-base reaction
- ☐ other

Part C

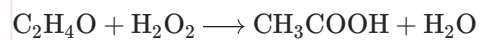
Reaction 3



- ☐ acid-base reaction
- ☐ other

Part D

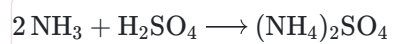
Reaction 4



- ☐ acid-base reaction
- ☐ other

Part E

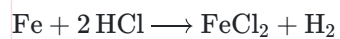
Reaction 5



- ☐ acid-base reaction
- ☐ other

Part F

Reaction 6



- ☐ acid-base reaction
- ☐ other

Part G

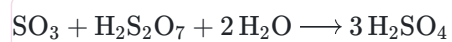
Reaction 7



- ☐ acid-base reaction
- ☐ other

Part H

Reaction 8



- ☐ acid-base reaction
- ☐ other

Question deck:

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